TRAINTY PROJECT 4

Hiring Process Analytics

Project Description: Analyze the company's hiring process data and draw meaningful insights from it and understanding trends such as the number of rejections, interviews, job types, and vacancies can provide valuable insights for the hiring department.

Analytical Tasks:

A. Hiring Analysis: The hiring process involves bringing new individuals into the organization for various roles.

Your Task: Determine the gender distribution of hires. How many males and females have been hired by the company?

B. Salary Analysis: The average salary is calculated by adding up the salaries of a group of employees and then dividing the total by the number of employees.

Your Task: What is the average salary offered by this company? Use Excel functions to calculate this.

C. Salary Distribution: Class intervals represent ranges of values, in this case, salary ranges. The class interval is the difference between the upper and lower limits of a class.

Your Task: Create class intervals for the salaries in the company. This will help you understand the salary distribution.

D. Departmental Analysis: Visualizing data through charts and plots is a crucial part of data analysis.

Your Task: Use a pie chart, bar graph, or any other suitable visualization to show the proportion of people working in different departments.

E. Position Tier Analysis: Different positions within a company often have different tiers or levels.

Your Task: Use a chart or graph to represent the different position tiers within the company. This will help you understand the distribution of positions across different tiers.

Tools Used: Excel and Trainity Portal

Handling Errors:

application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salary	H
249045	06-05-2014 13:26	Rejected	Female	Finance Department	c5	9009	
563405	06-05-2014 13:28	Rejected	Female	Finance Department	c5	40831	
810490	06-05-2014 13:30	Rejected	Female	Finance Department	c5	85140	
296052	29-05-2014 14:43	Hired	Female	Finance Department	c5	1141	
964704	29-05-2014 14:44	Hired	Female	Finance Department	c5	39485	
565183	06-05-2014 08:40	Rejected	Male	Marketing Department	b9	84675	
288171	06-05-2014 15:37	Hired	Female	Marketing Department	i4	33631	
499643	06-05-2014 09:26	Rejected	Female	Service Department	i4	45288	
610176	07-05-2014 14:06	Rejected	Male	Service Department	i4	46980	
346219	22-05-2014 10:21	Rejected	Female	Operations Department	i5	25621	
310984	07-05-2014 02:13	Rejected	Male	Service Department	i7	6472	
357945	07-05-2014 02:13	Rejected	Male	Service Department	i7	25239	
188820	07-05-2014 17:58	Rejected	Male	Sales Department	i4	94869	
193244	07-05-2014 10:53	Rejected	Male	Operations Department	i6	77157	
781620	07-05-2014 15:34	Rejected	Male	Production Department	i4	62894	
379557	07-05-2014 15:34	Hired	Female	Production Department	i4	61532	
674051	07-05-2014 15:35	Rejected	Female	Production Department	i4	81261	
928902	07-05-2014 16:26	Rejected	Female	Finance Department	b9	59644	
22155	07-05-2014 16:26	Rejected	Female	Finance Department	b9	46852	
572422	07-05-2014 08:07	Rejected	Male	Sales Department	i7	61488	
114584	07-05-2014 08:08	Rejected	Male	Sales Department	i7		
794226	07-05-2014 16:01	Rejected	Male	Service Department	i7	16090	
496573	07-05-2014 16:01	Rejected	Male	Service Department	i7	83364	
437695	13-05-2014 15:22	Rejected	Male	Service Department	i7	77517	
983577	08-05-2014 07:13	Rejected	Male	Service Department	b9	84746	
720850	08-05-2014 22:10	Rejected	Male	Sales Department	i6	80600	
453498	08-05-2014 03:12	Hired	Male	Sales Department	i4	62937	
550914	08-05-2014 03:15	Rejected	Female	Sales Department	i4	18921	
37906	08-05-2014 03:16	Hired	Female	Sales Department	i4	95603	
692080	08-05-2014 03:16	Hired	Male	Sales Department	i4	79230	
735087	08-05-2014 03:16	Rejected	Male	Sales Department	i4	56650	
855558	08-05-2014 11:52	Rejected	Female	Production Department	c5	75158	
480514	08-05-2014 19:13	Rejected	Female	Operations Department	i1	50125	

- **A blank cell can be seen in the highlighted row, since the candidate was rejected, we can delete the row.**

A	B	C	D	E	F	G	H
application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salary	
289907	01-05-2014 07:44	Hired	Male	Sales Department	-	85914	

- **There is a '- 'in the Post Name and upon investigating and finding a median of the post by comparing posts with similar salaries, the post name was changed to 'c9'.**

- **There were 27 duplicate application IDs found as well, which I removed making the total data count as 7140 rows.**

Removing Outliers:

- I found that there are outliers present in the salary column with 3 outliers at '200000', 300000', '40000' and these outliers were then replaced by the values '99967' as that was the highest matching value in the dataset.

A	B	C	D	E	F	G	H
application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salary	
795330	15-06-2014 09:45	Hired	Female	General Management	i4	400000	
874368	21-07-2014 15:39	Hired	Male	General Management	i7	300000	
649039	07-05-2014 10:48	Hired	Female	Service Department	b9	200000	
863024	01-08-2014 10:47	Hired	Male	Service Department	c8	99967	
913059	11-08-2014 09:29	Hired	Female	Service Department	c9	99953	
124098	23-06-2014 16:23	Hired	Male	Service Department	c9	99950	

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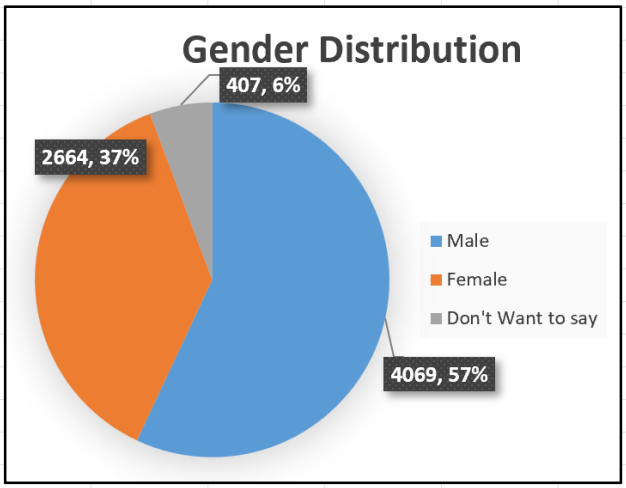
Task A: Determine the gender distribution of hires. How many males and females have been hired by the company?

Ans:

1. Gender Distribution Of Hires :

Gender	Count	Proportion
Male	4069	56.99%
Female	2664	37.31%
Don't Want to	407	5.70%



- The Gender distribution is as above with Males dominating at '56.99%' followed by Females at '37.31%'. The data was unclear for the rest of the employees as they did not share their gender.

Task B: What is the average salary offered by this company? Use Excel functions to calculate this.

Ans:

2. Average Salary offered by the company :

49925.91	is the average salary offered
	Calculated by AVERAGE Function applied throughout Salary Column
49649.45	is the average salary offered to hired people

- The average salary offered by the company is 49925.91
- The average salary offered to the people working in the company is 49649.45

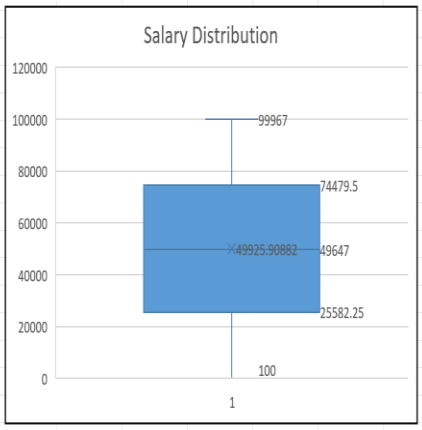
Task C: Create class intervals for the salaries in the company. This will help you understand the salary distribution.

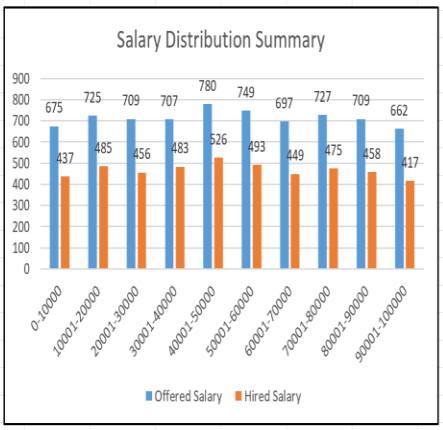
Ans:

3. Salary Distribution Of The Company :

Lowest Salary - 100
Highest Salary - 99967

Range	Offered Salary	Hired Salary
0-10000	675	437
10001-20000	725	485
20001-30000	709	456
30001-40000	707	483
40001-50000	780	526
50001-60000	749	493
60001-70000	697	449
70001-80000	727	475
80001-90000	709	458
90001-100000	662	417





- Since the salary ranged from 100 to 99967, I had divided the salary into 10 ranges with each range having an equal distribution of 9999.
- As you can see there is a significant difference between the count of offered salary and the count of hired people.
- The highest range is '40001-50000' followed closely by '50001-60000' and the lowest range was '90001-100000' followed closely by '0-10000'.

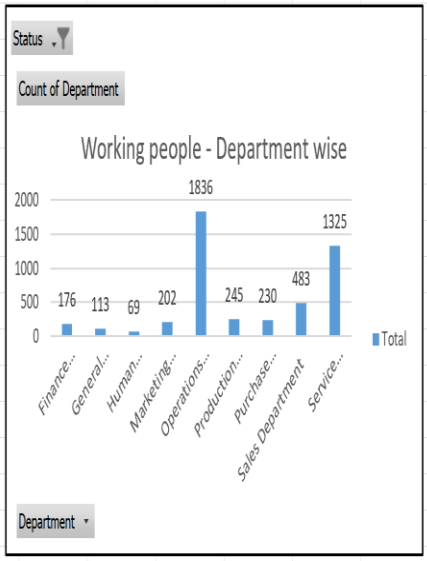
Task D: Use a pie chart, bar graph, or any other suitable visualization to show the proportion of people working in different departments.

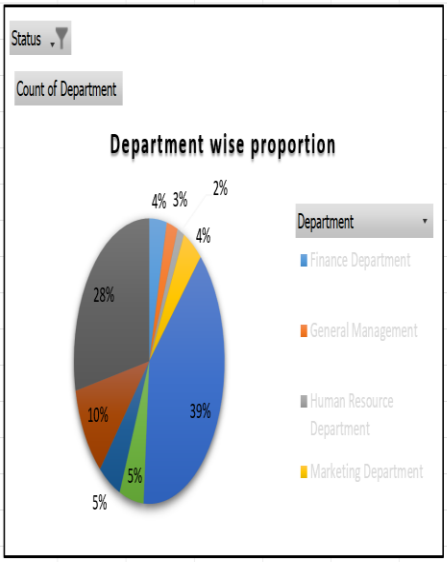
Ans:

4. Proportion of people working in different departments.

Status	Hired

Row Labels	Count of Department
Finance Depart	176
General Manag	113
Human Resour	69
Marketing Dep	202
Operations Dep	1836
Production Dep	245
Purchase Depa	230
Sales Departme	483
Service Departi	1325
Grand Total	**4679**





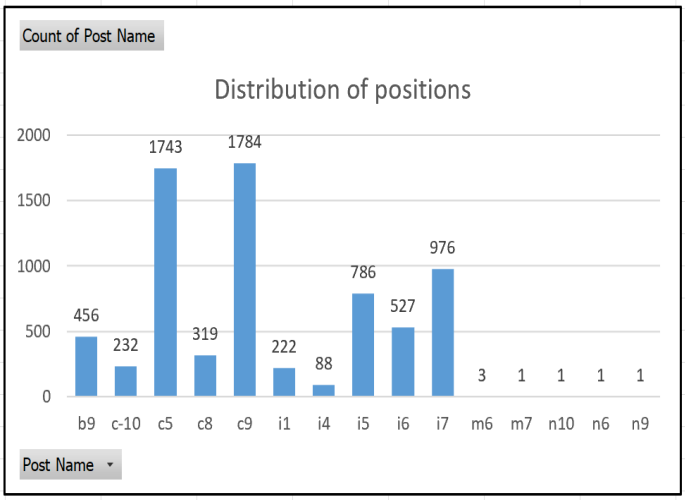
- The operations department has the highest number of employees actively working followed by the service department. Whereas the lowest number of employees actively working are for the human resources department followed by the marketing department.

Task E: Use a chart or graph to represent the different position tiers within the company. This will help you understand the distribution of positions across different tiers.

Ans:

5. Distribution of posiitons across the company

Post	Count of Post Name
b9	456
c-10	232
c5	1743
c8	319
c9	1784
i1	222
i4	88
i5	786
i6	527
i7	976
m6	3
m7	1
n10	1
n6	1
n9	1
Grand Total	**7140**



- The position with the highest number of employees is ‘c9’ followed by ‘c5’ whereas the position with the lowest number of employees is ‘m7’, ‘n10’, ‘n6’, ‘n9’ simultaneously having 1 employee each.

Result: Throught this project I understood the concept of using different formulas in excel, understanding which formulas to apply where and how to apply different conditions to formulas as well. I also understood the concept of pivot tables and pivot charts and how to apply filters to these charts as well. Alongside all these concepts, I understood formatting of charts as well along with choosing the right chart for the right type of data presentation which is very important for story-telling.

Excel Report:

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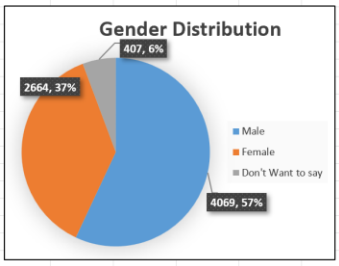

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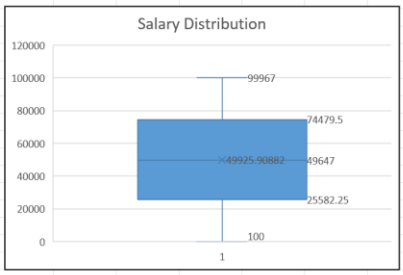

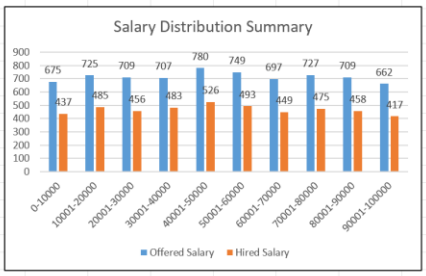

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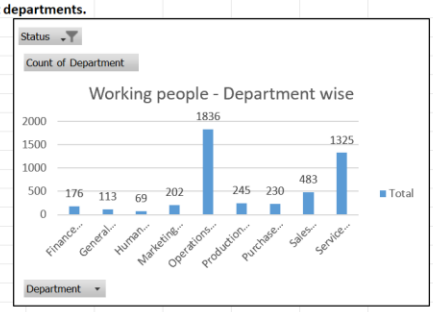

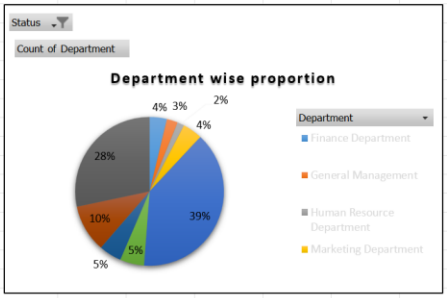

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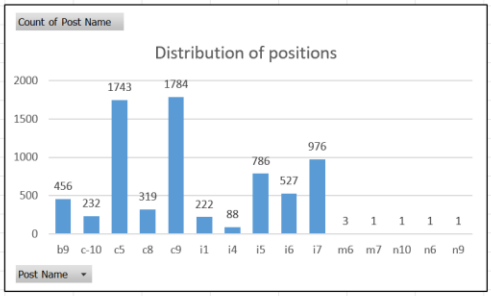

Link:
https://docs.google.com/spreadsheets/d/1LwO2rZUqXi0H8w1xO1Mz_VKLI3hobpcj/edit?usp=sharing&ouid=100878477496382804997&rtpof=true&sd=true